

“The Essentials of Computer Organization and Architecture”

Unit 2 – Exercises

//2.1. Perform the following base conversions using subtraction or division-remainder:

a) $458_{10} = \underline{\hspace{2cm}}_3$

b) $677_{10} = \underline{\hspace{2cm}}_5$

c) $1518_{10} = \underline{\hspace{2cm}}_7$

d) $4401_{10} = \underline{\hspace{2cm}}_9$

//2.2. Perform the following base conversions using subtraction or division-remainder:

a) $588_{10} = \underline{\hspace{2cm}}_3$

b) $2254_{10} = \underline{\hspace{2cm}}_5$

c) $652_{10} = \underline{\hspace{2cm}}_7$

d) $3104_{10} = \underline{\hspace{2cm}}_9$

2.3. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

- a) 26.78125
- b) 194.03125
- c) 298.796875
- d) 16.124023435

2.4. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

- a) 25.84375
- b) 57.55
- c) 80.90625
- d) 84.874023

2.5. Represent the following decimal numbers in binary using 8-bit signed magnitude, one's complement and two's complement:

- a) 77
- b) -42
- c) 119
- d) -107

2.6. Using a "word" of 3 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

- a) Signed magnitude
- b) One's complement
- c) Two's complement

2.7. Using a "word" of 4 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

- a) Signed magnitude
- b) One's complement
- c) Two's complement

2.16. Using signed-magnitude representation, and assuming a 4 bit representation, complete the following operations:

a) $+0 + (-0) =$

b) $(-0) + 0 =$

c) $0 + 0 =$

d) $(-0) + (-0) =$

2.18. If the floating-point number storage on a certain system has 1 sign bit, 3-bit exponent and 4-bit mantissa ...

... What is the largest positive and the smallest positive number that can be stored on this system if the storage is normalized?
(Assume no bits are implied, there is no biasing, **exponents use two's complement**, exponents of all 0's and all 1's aren't allowed.)

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

- a) Show how the computer would represent **100.0** and **0.25**.
- b) Show how the computer would add the two floating-point numbers of a) by changing one of the numbers so they are both expressed using the same power of 2.
- c) Show how the computer would represent the sum of b) using the given floating-point representation. What decimal value for the sum is the computer actually storing?

**2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$
Answer the following questions:**



1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {?} fractional part 0.{?}

$\Delta 1 = \{?\}$ (value lost in conversion)

B = integer part {?} parte fractional part 0.{?}

$\Delta 2 = \{?\}$ (value lost in conversion)

**2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$
Answer the following questions:**



**2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:**

2.a) Show how A and B would be represented in this model.

A = sign {?} | exponent {?} | significand {?}
 $\Delta 3 = \{?\}$ (value lost in representation)

B = sign {?} | exponent {?} | significand {?}
 $\Delta 4 = \{?\}$ (value lost in representation)

**2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$
Answer the following questions:**



2.b) Considering the sum $A + B$ in this model, complete the representation of the parts and the result of sum.

$$\begin{array}{r} \{\text{?}\}|\{\text{?}\}|\{\text{?}\} = A \\ + \{\text{?}\}|\{\text{?}\}|\{\text{?}\} = B \\ \hline \{\text{?}\}|\{\text{?}\}|\{\text{?}\} = A + B \end{array}$$

$\Delta 5 = \{\text{?}\}$ (value lost through denormalization of the smallest part)

$\Delta 6 = \{\text{?}\}$ (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

$\{\text{?}\}$

Confirmation: $A + B = 2.c) + \Sigma \Delta$

**2*.2. Solve exercise 2*.1 again for
A=101.0 and B=0.75**



**2*.3. Solve exercise 2*.1 again for
A=55.55 and B=7.777**



**2*.4. Solve exercise 2*.1 again for
A=64.0 and B=75.75**



**2*.5. Solve exercise 2*.1 again for
A=88.88 and B=9.999**



**2*.6. Solve exercise 2*.1 again for
A=66.66 and B=99.99**



**2*.7. Solve exercise 2*.1 again for
A=97.53 and B=35.79**



2.24. Conversions: ASCII and EBCDIC

- a) Knowing that ASCII code for the character A is $1000001_2 = 65_{10}$, what is the ASCII code for the character J?
- b) Knowing that EBCDIC code for the character A is $1100\ 0001_2$, what is the EBCDIC code for the character a ?

2.25. Assume a 24-bit word on a computer. How would the computer represent the string “295”, with 7-bit ASCII and a) even parity, or b) odd parity ?

0 NUL	16 DLE	32	48 0	64 @	80 P	96 `	112 p
1 SOH	17 DC1	33 !	49 1	65 A	81 Q	97 a	113 q
2 STX	18 DC2	34 "	50 2	66 B	82 R	98 b	114 r
3 ETX	19 DC3	35 #	51 3	67 C	83 S	99 c	115 s
4 EOT	20 DC4	36 \$	52 4	68 D	84 T	100 d	116 t
5 ENQ	21 NAK	37 %	53 5	69 E	85 U	101 e	117 u
6 ACK	22 SYN	38 &	54 6	70 F	86 V	102 f	118 v
7 BEL	23 ETB	39 '	55 7	71 G	87 W	103 g	119 w
8 BS	24 CAN	40 (	56 8	72 H	88 X	104 h	120 x
9 TAB	25 EM	41)	57 9	73 I	89 Y	105 i	121 y
10 LF	26 SUB	42 *	58 :	74 J	90 Z	106 j	122 z
11 VT	27 ESC	43 +	59 ;	75 K	91 [	107 k	123 {
12 FF	28 FS	44 ,	60 <	76 L	92 \	108 l	124
13 CR	29 GS	45 -	61 =	77 M	93]	109 m	125 }
14 SO	30 RS	46 .	62 >	78 N	94 ^	110 n	126 ~
15 SI	31 US	47 /	63 ?	79 O	95 _	111 o	127 DEL

Help:

2.26. Decode the following ASCII message, assuming 7-bit ASCII characters and no parity: 1001010 1001111 1001000 1001110 0100000 1000100 1001111 1000101

Help:

0 NUL	16 DLE	32	48 0	64 @	80 P	96 `	112 p
1 SOH	17 DC1	33 !	49 1	65 A	81 Q	97 a	113 q
2 STX	18 DC2	34 "	50 2	66 B	82 R	98 b	114 r
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5 ENQ	21 NAK	37 %	53 5	69 E	85 U	101 e	117 u
6 ACK	22 SYN	38 &	54 6	70 F	86 V	102 f	118 v
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15 SI	31 US	47 /	63 ?	79 O	95 _	111 o	127 DEL